

Build a dynamic scanner universe system that replaces the current hardcoded stock list. The scanner should be able to pull its universe from three sources: preset lists, custom watchlists, and dynamic screens. The user selects which universe to scan from a dropdown on the Scanner tab. The scanner scoring logic does not change regardless of which universe is selected.

LAYER 1 - PRESET UNIVERSES: Store these as JSON config files in the backend, not hardcoded in the scoring logic. Include these defaults at launch: "S&P 100" (the 100 largest US stocks by market cap), "S&P 500" (full S&P 500 membership), and "Nasdaq 100" (the 100 largest Nasdaq-listed stocks). These lists should be stored in a single JSON file like universes.json with each preset as a named array of ticker symbols. Update these lists manually once per quarter or when index membership changes. No API needed for this layer.

LAYER 2 - CUSTOM WATCHLISTS: Build a simple watchlist management UI accessible from the Watchlist tab or a Settings page. The user can create named watchlists (for example "My Earnings Plays" or "Sector Rotation" or "Dividend Names"), add tickers by typing or searching, remove tickers, rename or delete watchlists. Store watchlists in the database tied to the user account. Each watchlist is just a name and an array of ticker symbols. These custom watchlists appear in the scanner universe dropdown alongside the presets.

LAYER 3 - DYNAMIC SCREENS: This is the most important layer. A dynamic screen is a saved set of filters that generates a fresh ticker list every time it runs. The user creates a screen by setting filter criteria. The system runs the screen against a market data API and returns all tickers that match. The resulting ticker list is then fed to the scanner for scoring.

FILTER CRITERIA FOR DYNAMIC SCREENS: Each screen has a name and a set of filters. All filters are optional. If a filter is not set it is ignored (no constraint). The available filters are: minimum market cap in dollars (example: 10000000000 for 10 billion), maximum market cap in dollars (for small cap screens), minimum average daily volume in shares over the last 20 trading days (example: 2000000 for 2 million), minimum stock price (example: 20 to exclude penny stocks), maximum stock price (optional), sector inclusion list (example: Technology, Healthcare, Financial Services), sector exclusion list (example: exclude Utilities and Real Estate), exchange filter (NYSE, NASDAQ, or both), minimum options volume in contracts per day (example: 500 to ensure liquid options), and country filter (default US only). The screen also has a max results cap (default 200) to prevent the scanner from being overwhelmed.

SCREEN EXECUTION: When a dynamic screen runs, the backend calls a market data API with the filter criteria and receives back a list of matching tickers. The screen runs automatically once per day at 8:00 AM ET (pre-market) and caches the results.

The cached results are used for all scanner runs that day. The user can also manually refresh a screen from the UI to get an updated list mid-day. Store the cached results with a timestamp so the UI can show "Universe: Dynamic Large Cap (generated 8:00 AM, 347 stocks)".

DATA SOURCE FOR SCREENING: Use FMP (Financial Modeling Prep) stock screener API endpoint for dynamic screens. FMP's screener supports filtering by market cap, price, volume, sector, exchange, and other criteria. The API call looks like: <https://financialmodelingprep.com/api/v3/stock-screener> with query parameters for marketCapMoreThan, marketCapLessThan, volumeMoreThan, priceMoreThan, priceLessThan, sector, exchange, country, and limit. FMP is already being evaluated for earnings dates so this can share the same API key and subscription. If FMP is not yet connected, the dynamic screen layer can be built with the API integration point stubbed out and the preset and custom watchlist layers will still work independently.

DEFAULT DYNAMIC SCREENS: Create these pre-built dynamic screens that the user can modify or duplicate: "Large Cap Liquid" with filters market cap above 10 billion and average volume above 2 million and options volume above 500 and price above 20 (this replaces the old hardcoded Top 100 list with a dynamic equivalent that adapts to market conditions), "Mid Cap Movers" with filters market cap between 2 billion and 10 billion and average volume above 1 million and price above 10, "High IV Opportunity" with filters market cap above 5 billion and average volume above 1 million (this list gets further filtered by the scanner's IV Rank scoring so it does not need an IV filter in the screen itself).

SCANNER UNIVERSE DROPDOWN: On the Scanner tab, add a dropdown selector above the scanner results. The dropdown shows all available universes grouped by type: a Presets section showing S&P 100, S&P 500, Nasdaq 100; a Dynamic section showing Large Cap Liquid (347 stocks), Mid Cap Movers (215 stocks), High IV Opportunity (280 stocks); a Custom section showing any user-created watchlists. The count in parentheses shows how many tickers are in that universe. The default selection is "Large Cap Liquid" (the dynamic replacement for the old hardcoded list). The selected universe persists across sessions. When the user switches universes, the scanner re-runs automatically with the new list.

SCREEN BUILDER UI: Build a screen creation and editing interface accessible from a "Create Screen" button in the scanner dropdown or from a Settings page. The interface shows each filter as a labeled row with an input field. Market cap fields use a dropdown for common values (1B, 2B, 5B, 10B, 50B, 100B) or allow custom entry. Volume field is a number input. Sector filter is a multi-select checkbox list of GICS sectors. Price fields are number inputs. Each filter has a toggle to enable or disable it. A "Preview" button runs the screen and shows how many stocks match before saving. A "Save Screen" button saves it with a name. Saved screens appear in the scanner dropdown immediately.

WATCHLIST MANAGEMENT UI: On the Watchlist tab or accessible from Settings, show

the list of custom watchlists. Each watchlist shows its name, ticker count, and a list of tickers with remove buttons. An "Add Ticker" input with autocomplete search lets the user type a symbol or company name and add it. A "Create New Watchlist" button creates a blank watchlist with a name prompt. Watchlists can be renamed or deleted. A special "Favorites" watchlist is pre-created and cannot be deleted.

INTEGRATION WITH SCANNER: The scanner scoring engine receives an array of ticker symbols as input regardless of where they came from (preset, custom, or dynamic screen). The scoring logic does not change. The scanner does not know or care whether the universe was static or dynamically generated. This keeps the architecture clean. The universe selection is purely a data input layer.

VOLUME SCORING FIX: Currently the scanner scores volume confirmation by comparing partial-day intraday volume against a 20-day average using the elapsed fraction projection that was fixed earlier today. When the universe changes, the volume data for new tickers needs to be fetched. Make sure the scanner fetches 20-day average volume for whatever tickers are in the selected universe, not just the old hardcoded list. This may require a batch API call to Schwab or FMP for historical volume data on first use of a new universe.

PERSISTENCE: Preset universe JSON files are stored in the backend repo. Custom watchlists are stored in the database per user. Dynamic screen definitions (filter criteria) are stored in the database per user. Dynamic screen cached results (the generated ticker list) are stored in memory or a lightweight cache with a timestamp, regenerated daily at 8:00 AM ET. Scanner universe selection (which universe is currently active) is stored as a user preference in the database.